Edge of Stability Occurs in Minimax Optimization

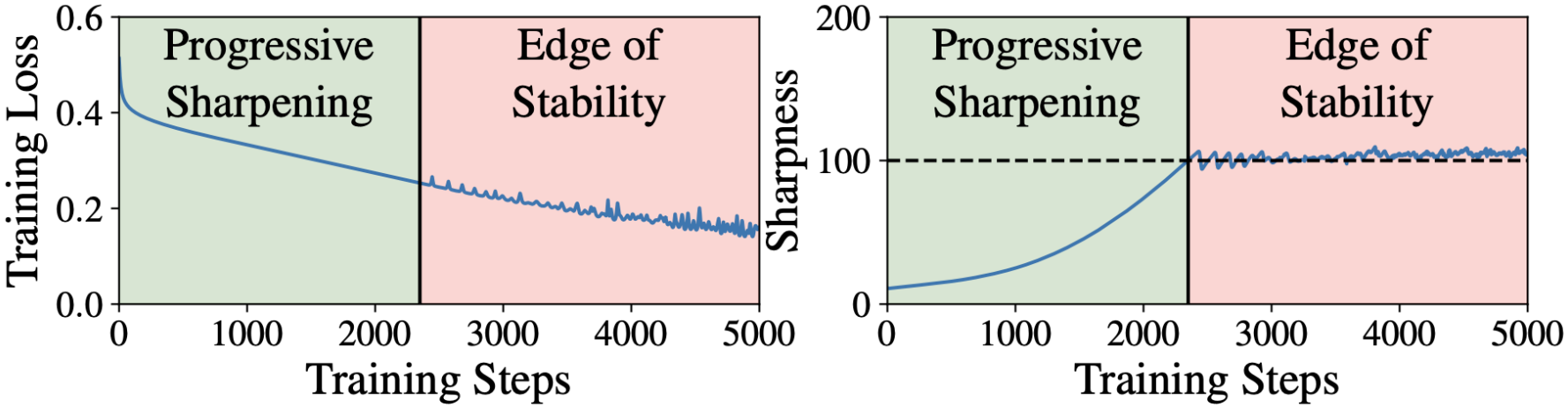
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Learning Theory (LeT) Workshop

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Edge of Stability in Neural Network Training



- Neural network training exhibits unexpected stability beyond what classical optimization theory predicts.

Cohen, Kaur, Li, Kolter, and Talwalkar, Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, ICLR, 2021.

Gradient Descent in Minimization

- Minimization

$$\min_z g(z).$$

- Gradient Descent (GD)

$$z_{t+1} = z_t - \eta \nabla g(z_t).$$

- The stability of GD is closely related to the largest Hessian eigenvalue

$$S(z) := \lambda_{\max}\big(\nabla^2 g(z)\big).$$

Stability of Gradient Descent (GD)

- Assume that g is L-smooth. For twice-differentiable g, this implies

$$\|\nabla^2 g(z)\| \leq L, \qquad S(z) \leq L.$$

- Then GD satisfies the descent inequality

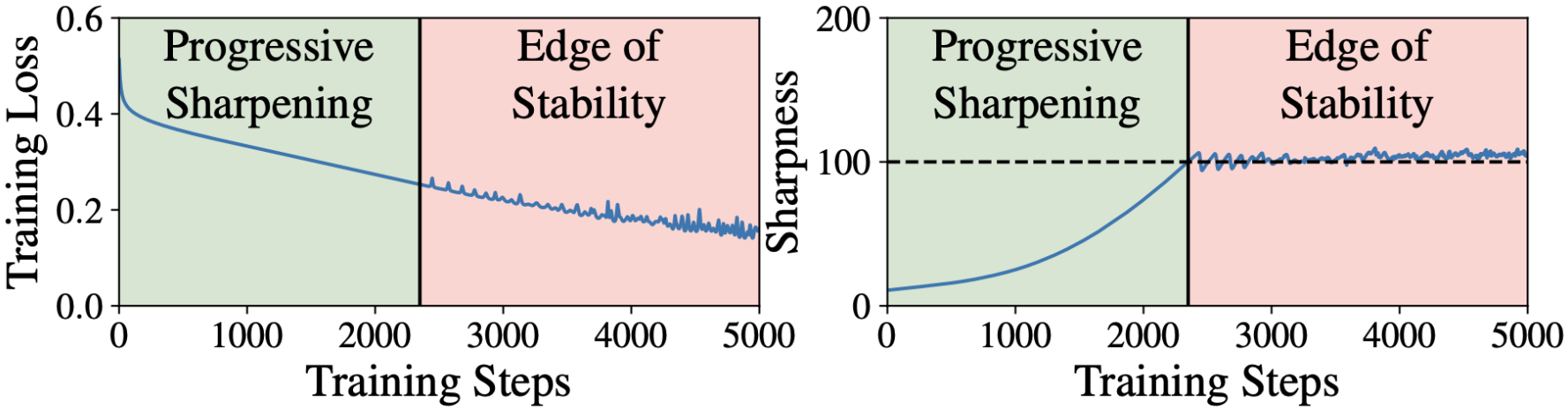
$$g(z_{t+1}) \leq g(z_t) - \eta \left(1 - \frac{\eta L}{2}\right) \|\nabla g(z_t)\|^2.$$

- Hence, descent is guaranteed when

$$L < \frac{2}{\eta}.$$

- But in neural network training, $S(z_t)$ evolves during training and approaches $\frac{2}{\eta}$.

Edge of Stability in Minimization



- **Progressive Sharpening**: $S(z_t)$ progressively increases toward $\frac{2}{\eta}$.
- **Edge of Stability (EoS)**: After reaching the threshold, $S(z_t)$ remains near it while the loss continues to decrease.
- Why does training not simply diverge once it reaches this threshold?

Cohen, Kaur, Li, Kolter, and Talwalkar, Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, ICLR, 2021.

Self-Stabilization of GD

- Damian et al. (2023) explain the EoS behavior of GD through a self-stabilization mechanism.
- Let $u(z)$ be the top Hessian eigenvector.
- Once the dynamics become unstable along $u(z)$, the growing unstable component induces the second-order term in the GD update

$$-D^2(\nabla g)(z)[u(z), u(z)] = -\nabla S(z),$$

reducing $S(z)$ and pushing GD back toward stability.

Damian, Nichani, and Lee, Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability, ICLR, 2023.

Minimax Optimization

- Minimax optimization

$$\min_x \max_y f(x, y).$$

  This arises in game-theoretic learning problems.

- Define the saddle-gradient operator

$$F(z) = \begin{bmatrix} \nabla_x f(x, y) \\ -\nabla_y f(x, y) \end{bmatrix}, \qquad z = (x, y).$$

- Gradient Descent Ascent (GDA),

$$z_{t+1} = z_t - \eta F(z_t),$$

  and GDmax both diverge even on bilinear problems, *e.g.*, $f(x, y) = xy$.

Extragradient and Optimistic GDA

- Extragradient (EG)[†]: $z_{t+1} = z_t - \eta F(z_t - \eta F(z_t))$
- Optimistic GDA (OGDA)[*]: $z_{t+1} = z_t - \eta \left(2F(z_t) - F(z_{t-1})\right)$
- Both EG and OGDA are guaranteed to converge for broad classes of minimax problems.
- But little is known about their stability behavior in practical minimax training with neural networks.

[†] Korpelevich, Extragradient Method for Finding Saddle Points and Other Problems, Ekonomika i Matematicheskie Metody, 1976.

[*] Popov, A Modification of the Arrow-Hurwicz Method for Search of Saddle Points, Mathematical Notes of the Academy of Sciences of the USSR, 1980.

Overview

1. How should we define the Edge of Stability in minimax optimization?
2. Does the Edge of Stability occur in minimax optimization?
3. Can minimax methods also self-stabilize at the edge?

Joint work with Yejun Kim and Dongchan Shin

Stability of GD: Nonexpansiveness

- View GD as a fixed-point iteration

$$z_{t+1} = (I - \eta \nabla g)(z_t).$$

- For convex L-smooth g, its gradient ∇g is $1/L$-cocoercive, and hence

$$\begin{aligned}\|(I - \eta \nabla g)(z) - (I - \eta \nabla g)(z')\|^2 \\ \leq \|z - z'\|^2 - \eta \left(\frac{2}{L} - \eta\right) \|\nabla g(z) - \nabla g(z')\|^2.\end{aligned}$$

- Therefore, $I - \eta \nabla g$ is *nonexpansive* when

$$0 \leq \eta \leq \frac{2}{L},$$

which recovers the familiar stability threshold $L = \frac{2}{\eta}$.

Stability of Minimax Methods: Nonexpansiveness

- More generally, consider a fixed-point iteration

$$z_{t+1} = (I - \eta T)(z_t) = z_t - \eta T(z_t).$$

- A standard notion of stability is the *nonexpansiveness* of $I - \eta T$.

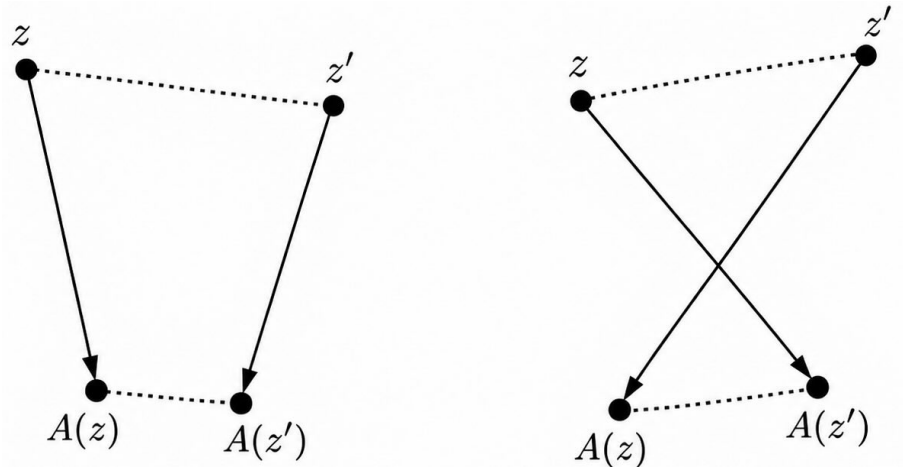
- We consider the following stronger form

$$\|(I - \eta T)(z) - (I - \eta T)(z')\|^2 \leq \|z - z'\|^2 - \alpha \|T(z) - T(z')\|^2, \quad (\star)$$

  which reduces to nonexpansiveness when $\alpha = 0$.

Stability in the Complex Plane

- To represent stability geometrically and relate it to eigenvalues, we use the Scaled Relative Graph (SRG).



- The SRG of $A = I - \eta T$,

$$\mathcal{G}(A) = \left\{ \frac{\|A(z) - A(z')\|}{\|z - z'\|} e^{\pm i \angle (A(z) - A(z'),\, z - z')} \,:\, z \neq z' \right\},$$

lies in the unit disk if and only if A is *nonexpansive*.

Ryu, Hannah, and Yin, Scaled Relative Graphs: Nonexpansive Operators via 2D Euclidean Geometry, Mathematical Programming, 2022.

Stability in the Complex Plane

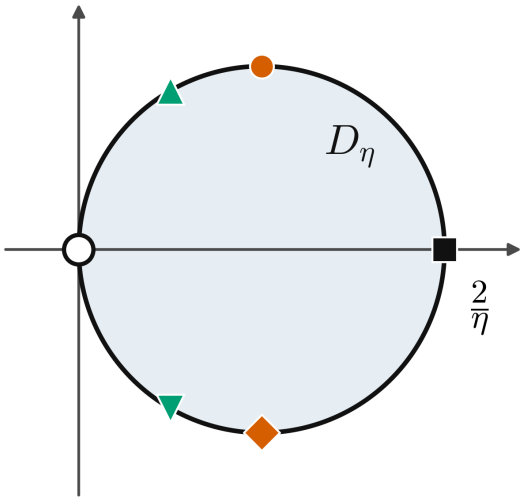
- We can equivalently express *nonexpansiveness* in terms of $\mathcal{G}(T)$.
- For any $\ell > 0$, the inequality $(\star)$ holds with $\alpha = \eta(\ell - \eta)$ if and only if

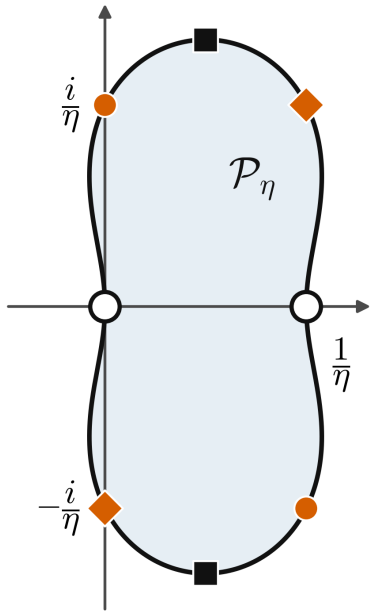
$$\mathcal{G}(T) \subset \mathcal{D}_\ell := \left\{ \omega \in \mathbb{C} : \left| \omega - \frac{1}{\ell} \right| \leq \frac{1}{\ell} \right\}.$$

- In particular, at $\ell = \eta$, we have $\alpha = 0$, so $I - \eta T$ is nonexpansive if and only if $\mathcal{G}(T) \subset \mathcal{D}_\eta$.
- Moreover, $\mathcal{G}(T) \subset \mathcal{D}_\eta$ is directly related to the eigenvalues of the Jacobian DT.

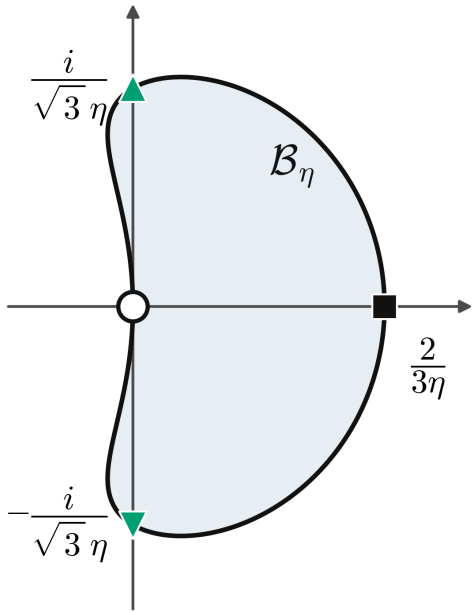
Method-Dependent Stability Regions

- $\mathcal{G}(T) \subset \mathcal{D}_\eta$ implies $\mathrm{eig}(DT(z)) \subset \mathcal{D}_\eta$ for all z.
- The converse also holds when $DT(z)$ is *normal* for all z.
- Stability regions can also be expressed in terms of the eigenvalues of DF.





EG: $\mathcal{P}_\eta$



OGDA: $\mathcal{B}_\eta$

Minimax EoS

- **Progressive sharpening**: the magnitudes of some eigenvalues of $DF(z_t)$ progressively increase during training.

- Let $S(z_t)$ denote the eigenvalue of $DF(z_t)$ that reaches the stability boundary.

- **Edge of Stability**: $S(z_t)$ reaches the boundary

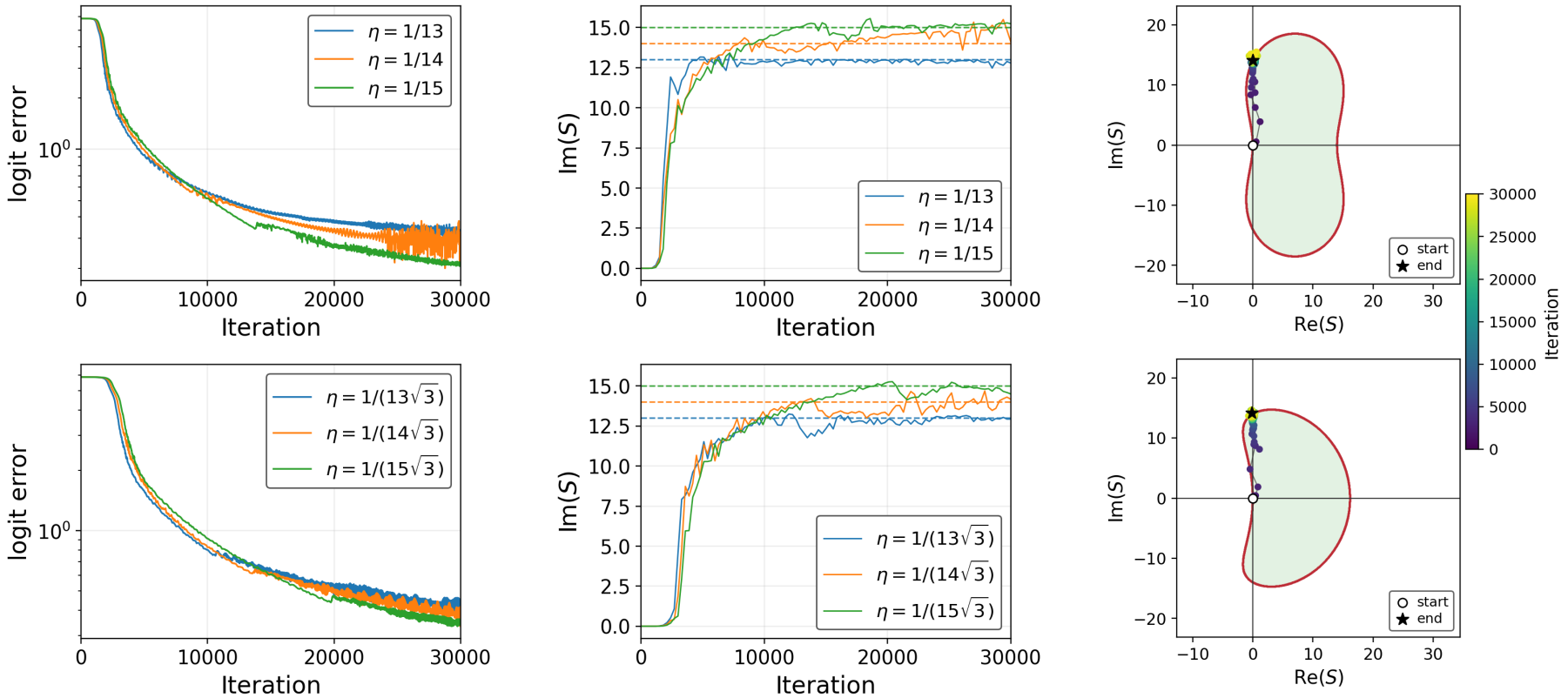
$$\partial\mathcal{P}_\eta \quad \text{or} \quad \partial\mathcal{B}_\eta$$

and remains near it while training continues to make progress.

EoS Occurs in Two-Player Zero-Sum Games

(1) Two-stage RPS game (가위바위보 하나 빼기)

Each player's strategy is parameterized by a neural network.

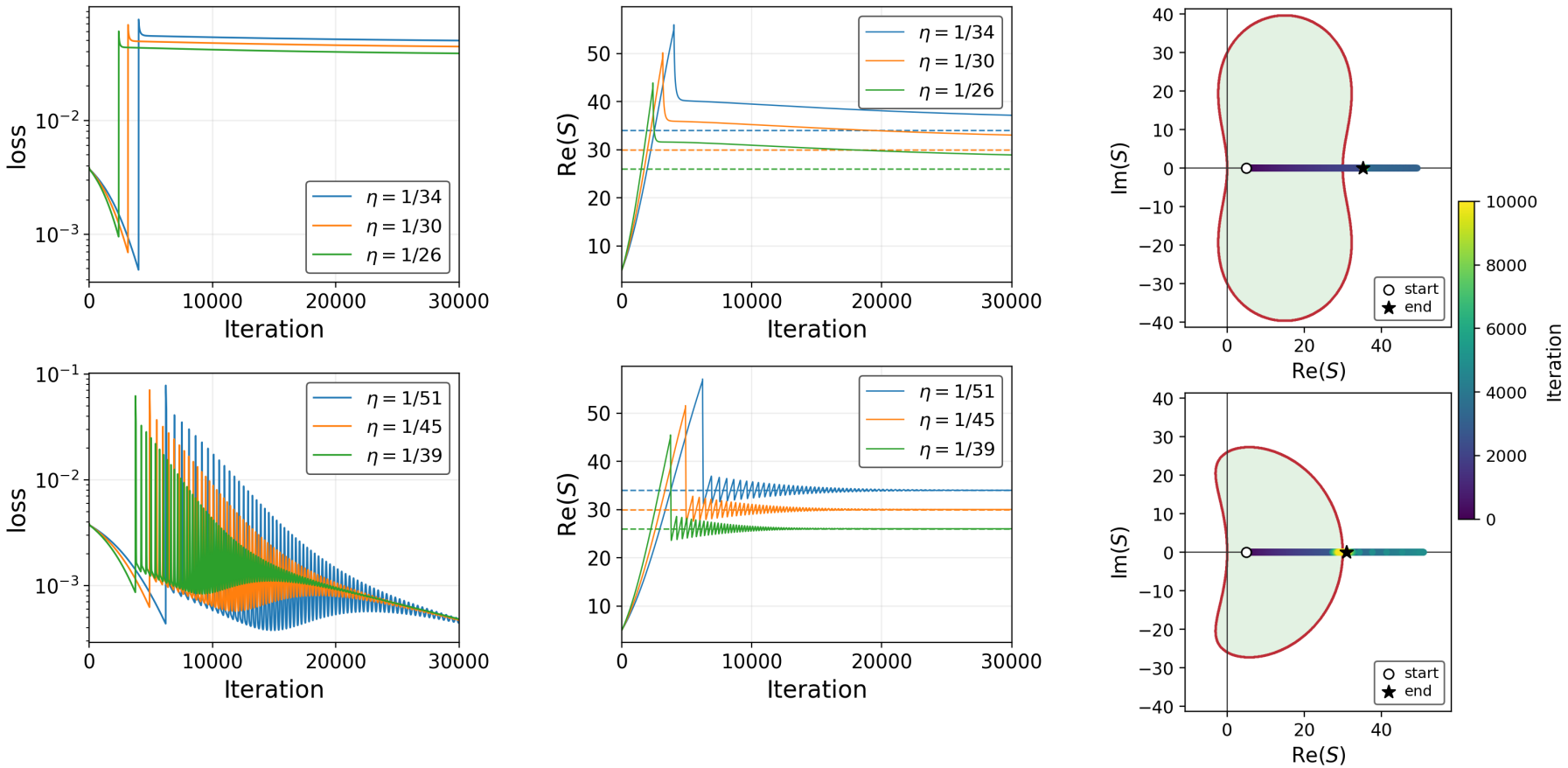


- We observed similar behavior for the pursuit-evasion game (a multi-stage Markov game).

Near Real Axis: OGDA Exhibits EoS, While EG Does Not

(2) Simple minimization problem[‡]

A notable difference appears when $S(z_t)$ lies near the real axis.



- We observed similar behavior for another pursuit-evasion game and neural optimal transport, when $S(z_t)$ is near, but not exactly on, the real axis.

[‡] Liu, Zhang, Du, and Zhao, A Minimalist Example of Edge-of-Stability and Progressive Sharpening, NeurIPS, 2025.

From Stability Boundary to Self-Stabilization

- Let $\Lambda(z)$ be the eigenvalue of DT associated with $S(z)$.
- Define

$$\phi(z) := \frac{|1-\eta\Lambda(z)|^2 - 1}{2\eta},$$

  so that

$$\phi(z) > 0 \quad \iff \quad \Lambda(z) \notin \mathcal{D}_\eta.$$

- For GD in minimization, $\Lambda = S \in \mathbb{R}$ and thus when $S \approx 2/\eta$,

$$\nabla\phi \approx \nabla S.$$

Self-Stabilization Property

- We define the self-stabilization coefficient

$$\beta(z) := \nabla\phi(z) \cdot D^2T(z)[u(z), \bar{u}(z)],$$

and say that the self-stabilization property holds when

$$\beta(z) > 0.$$

This is guaranteed for GD, but not in general.

- Once the dynamics become unstable along the eigendirections $u(z)$ and $\bar{u}(z)$, the growing unstable component induces the second-order term

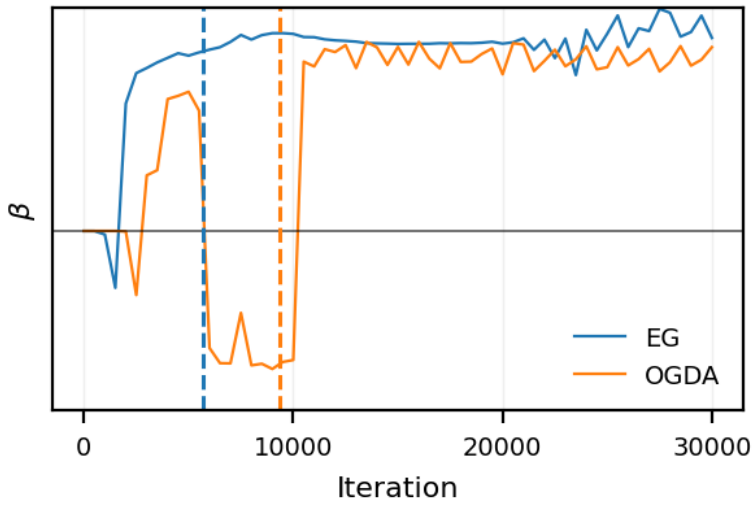
$$-D^2T(z)[u(z), \bar{u}(z)]$$

in the update.

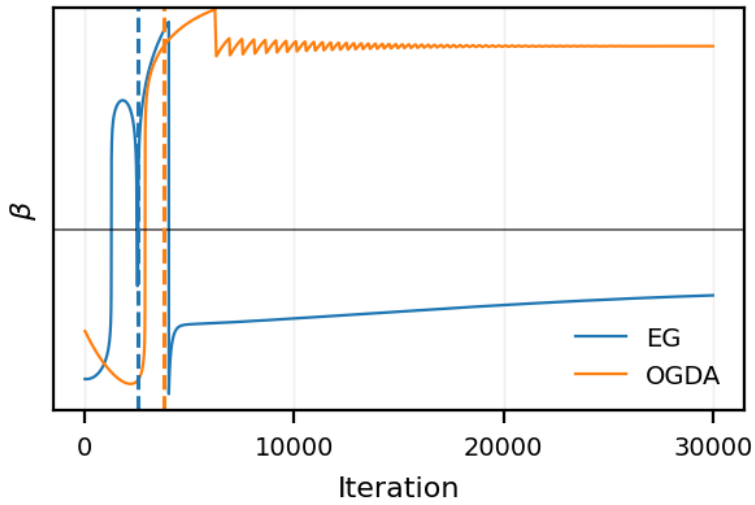
- If $\beta(z) > 0$, this term has a component that decreases ϕ, pushing the dynamics back toward $\mathcal{D}_\eta$.

Self-Stabilization and EoS

- The sign of β is closely associated with the emergence of EoS behavior.



(1) Two-stage RPS



(2) Simple minimization

The dashed line indicates the first iteration at which $\Lambda(z_t) \notin \mathcal{D}_\eta$.

- For EG near the real axis,

$$S(z_t) \to \frac{1}{\eta} \quad \Longrightarrow \quad \Lambda(z_t) \approx S(z_t) - \eta S(z_t)^2 \to 0,$$

which makes the condition for $\beta(z_t) < 0$ easier to satisfy.

Takeaways

- The notion of EoS can be generalized to minimax optimization.
- EoS occurs for both EG and OGDA, but their stability behavior can differ: near the real axis, OGDA exhibits EoS while EG does not.
- The self-stabilization principle of GD extends to minimax methods and helps explain when EoS emerges or fails.